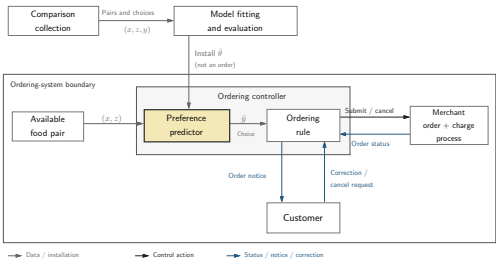


Same predictor, changed conditions



Hold the predictor fixed. Which changes in deployment can alter its performance?



Roadmap

Distribution shift: which mixtures?

Adversarial inputs: which transformations?

Calibration: which probability-outcome relationships?

The uncertainty set

Which deployment distributions does the claim cover?

- Group mixtures** Allow familiar groups to occur in different proportions.
- Adversarial inputs** Allow targeted changes to individual inputs.
- Calibration** Require reported probabilities to agree with outcome frequencies.

Assuming calibration restricts the distributions covered. It does not make arbitrary shifts safe.

A shift can change the weighting of familiar groups

Freshness matched and unmatched categories:

Group	Group error	Source weight	Shifted weight
Unmatched Category	2%	95%	50%
Matched Category	20%	5%	50%
Aggregate error		2.9%	11%

Specify the robustness claim

Fix $\hat{\theta}$. Evaluate prediction error against the intended-choice labels under deployment distribution S' .

$$R_{S'}(\hat{\theta}) = \mathbb{E}_{(x,z,y) \sim S'}[\ell_{01}(\hat{y}_{\hat{\theta}}(x,z), y)]$$

For every shift in the specified family:

$$\forall S' \in \mathcal{U} : \quad R_{S'}(\hat{\theta}) \leq \tau.$$

Better in both groups, worse overall

Suppose you do stratified evals + DRO. Deployment distribution changes still:

Group	Error before	Error after	Weight: before / after
Unmatched Category	2%	1%	95% / 50%
Matched Category	20%	12%	5% / 50%
Aggregate	2.9%	6.5%	

Simpson's paradox: different mixtures reverse the aggregate comparison.

Robust optimization changes the selection objective

Average-risk objective

$$\min_{\theta} R_S(\theta)$$

Distributionally robust objective

$$\min_{\theta} \sup_{S' \in \mathcal{U}} R_{S'}(\theta)$$

Specifying $\mathcal{U}$ defines which changes the robustness claim covers.

For group mixtures, the worst group determines risk

Let S_g be fixed within-group distributions. Allow every mixture:

$$\mathcal{U}_{\text{group}} = \left\{ \sum_{g=1}^m q_g S_g : q_g \geq 0, \sum_{g=1}^m q_g = 1 \right\}$$

$$R_{\sum_g q_g S_g}(\hat{\theta}) = \sum_g q_g R_{S_g}(\hat{\theta})$$

$$\sup_{S' \in \mathcal{U}_{\text{group}}} R_{S'}(\hat{\theta}) = \max_g R_{S_g}(\hat{\theta})$$

What evidence covers every mixture?

Evidence	Simultaneously valid upper bounds establish $R_{S_g}(\hat{\theta}) \leq \tau$ for every group.
Scope	The predictor, within-group distributions, labels, and evaluation loss match the claim.
Estimation challenge	Small groups can have substantial uncertainty even when observed error is low.

Stratified evaluation estimates group risk; group DRO uses group losses to select a predictor.

Gender Shades: disaggregate performance

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0%	79.2%	100%	98.3%	20.8%
 FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
 IBM	88.0%	65.3%	99.7%	92.9%	34.4%

Commercial binary gender classification, evaluated across annotated gender and skin-tone groups.

Buolamwini and Gebru, Gender Shades, 2018



Roadmap

Distribution shift: which mixtures?

Adversarial inputs: which transformations?

Calibration: which probability-outcome relationships?

From reweighting to targeted inputs

Reweightings changes which cases arrive. An adversary can change the case itself to exploit the predictor.

Ordering example Edit a food photograph so the fixed predictor selects the unwanted substitute.

Robustness question Does the prediction remain correct under every allowed, label-preserving edit?

Define the allowed transformations before searching for an attack.

Specify what the adversary may change

Here x is the complete predictor input: the full comparison in the food example.

$$\tilde{x} = x + \eta, \quad \|\eta\|_{\infty} \leq \epsilon$$

Permitted changes Each coordinate can vary by at most ϵ , subject to valid-input constraints.

Intended label Justify preservation of food identity, freshness, and the customer's intended choice.

Small coordinate changes can accumulate

For the linear predictor $f_w(x) = \text{sign}(w^{\top}x)$:

$$w^{\top}\tilde{x} = w^{\top}x + w^{\top}\eta$$

$$\eta = \epsilon \text{sign}(w)$$

$$w^{\top}\eta = \epsilon \sum_{i=1}^n |w_i| = \epsilon \|w\|_1$$

Goodfellow et al., 2014

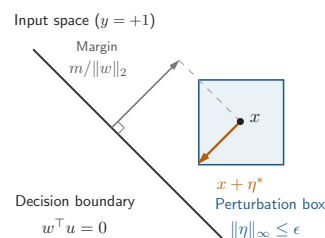
Can the perturbation cross the decision boundary?

For a correctly classified input, the signed margin is $m = y w^{\top}x > 0$.

$$y w^{\top}(x + \eta) \geq m - \epsilon \|w\|_1$$

Sufficient for unchanged classification:

$$m > \epsilon \|w\|_1$$



A failed search is not a certificate

Successful attack A permitted, label-preserving counterexample refutes the claimed invariance for that instance.

Unsuccessful search The tested procedure did not find a counterexample within its search budget.

Certificate A justified bound or verification covers every permitted perturbation in its stated scope.



Roadmap

Distribution shift: which mixtures?

Adversarial inputs: which transformations?

Calibration: which probability-outcome relationships?

What does a reported probability mean?

If a predictor reports 0.9, does label +1 occur 90% of the time among those cases?

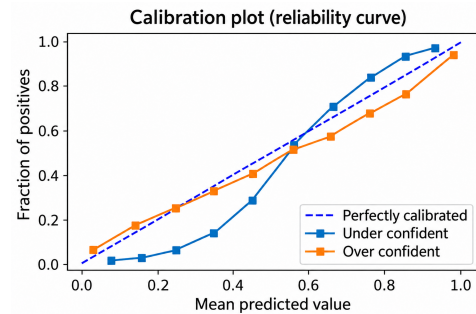
Calibration under S :

$$\Pr_S(Y = +1 \mid \hat{p}(X) = p) = p$$

For a binary predictor choosing the more probable label:

$$C(X) = \max\{\hat{p}(X), 1 - \hat{p}(X)\}$$

Reading a reliability diagram



Horizontal: mean predicted probability of label +1. Vertical: observed fraction of label +1.

The diagonal represents agreement. Check the counts, uncertainty, and population behind each bin.

Calibration does not imply discrimination

Illustrative population: label +1 occurs in 70% of cases.

Predictor	Reports $\hat{p}(X) = 0.7$ for every input.
Calibration	Among cases receiving 0.7, label +1 occurs 70% of the time.

Perfectly calibrated, but unable to distinguish individual cases. Always selecting +1 gives 30% error.

Calibration restricts the uncertainty set

For a fixed probability predictor, define the calibrated distributions:

$$\mathcal{U}_{\text{cal}} = \{S' : \Pr_{S'}(Y = +1 \mid \hat{p}(X) = p) = p\}$$

A claim assuming deployment calibration covers:

$$S' \in \mathcal{U} \cap \mathcal{U}_{\text{cal}}$$

Calibration on S does not show that deployment lies in this smaller set.

Ovadia et al., NeurIPS 2019

Reweighting can break pooled calibration

A fixed predictor reports $\hat{p}(X) = 0.9$ for every case.

Group	Frequency of +1	Source weight	Shifted weight
A	100%	50%	20%
B	80%	50%	80%
Pooled		90%	84%

Calibrated at one mixture does not mean calibrated at every mixture.

Using calibration: a proposed confidence gate

Automatically submit only if selected-label confidence $C > t$.



Assume calibration. With selected-label calibration under deployment S' :

$$\Pr_{S'}(\hat{y} \neq Y \mid C > t) = \mathbb{E}_{S'}[1 - C \mid C > t] \leq 1 - t$$

Routing changes the labeled distribution

Fix a calibrated predictor and source S . If every requested confirmation yields a faithful label:

$$S_{\text{conf}} = S(\cdot \mid C \leq t)$$

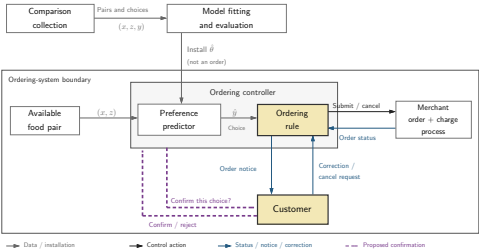
- Thresholding alone** Exact calibration survives at retained confidence levels. There are no labels for $C > t$.
- Selective responses** If only customers given a wrong prediction respond at $C = 0.8 \leq t$, observed accuracy is 0%, not 80%.

Recap: does the claim cover deployment?

- Group mixtures** All mixtures: worst risk = worst-group risk. Changing weights can reverse the aggregate comparison.
- Adversarial edits** A margin bound can certify allowed edits; a failed attack search is not a certificate.
- Calibration and routing** Assuming calibration restricts covered shifts. Routing changes label coverage; response selection can break calibration.

Specify the uncertainty set. Match evidence to the claim. Check the remaining system controls.

Which causal step does the intervention change?



Changing inputs, improving the predictor, and enforcing confirmation constrain different causal steps.

Discussion: requirements, controls, and evidence

The ordering platform is considering an operating change. For example, label-preserving normalization of food photographs or requiring explicit customer confirmation. Pick one change and retain it for the questions below.

Discussion: changed operating conditions

Specify one changed operating condition relevant to the ordering harm scenario. What remains fixed, including the customer's intended choice?

Discussion: model-level requirements

State the corresponding model-level requirement. What evidence would establish it, and which conditions would remain outside its scope?

Discussion: causal steps and remaining evidence

Identify the causal step that meeting this requirement would constrain. What further evidence is needed about ordering and feedback?

Discussion: confirmation policies

Compare universal confirmation with confirmation only for low-confidence predictions. Which routes to unwanted charges remain under each proposal, and what evidence would test the relevant controls?

Next: data as specification

Picking a population, uncertainty set, subgroups, labels, or permitted transformations is specification work.

- Population and groups** Which conditions and comparisons does the claim cover?
- Labels and transformations** Whose intended choice is recorded, and which changes preserve it?
- Tolerance and controls** What error is acceptable, and what prevents an error from becoming an unwanted charge?